BME403- MANUFACTURING PROCESSES

CO-1 : Students will learn the various conventional manufacturing processes / casting and forming processes. **K2**
CO-2 : Students will understand the concepts of metal cutting and CNC machining. **K2**
CO-3 : Students will comprehend the knowledge of grinding and super finishing processes. **K2**
CO-4 : Students will understand the concepts of metal joining processes. **K2**
CO-5 : Students will learn the concepts of unconventional machining processes. **K2**

UNIT-I (08 Lecture)
Conventional Manufacturing processes:
Casting and moulding: Metal casting processes and equipment, Types of Patterns and Pattern Allowance, Heat transfer and solidification, shrinkage, riser design, casting defects and residual stresses. Introduction to bulk and sheet metal forming, plastic deformation and yield criteria; fundamentals of hot and cold working processes; load estimation for bulk forming (forging, rolling, extrusion, drawing) and sheet forming (shearing, deep drawing, bending), principles of powder metallurgy.

UNIT-II (08 Lecture)
Metal cutting:
Single and multi-point cutting; Orthogonal cutting, various force components: Chip formation, Tool wear and tool life, Surface finish and integrity, Machinability, cutting tool materials, cutting fluids, Coating; Turning, Drilling, Milling and finishing processes,
Introduction to CNC machining, Additive manufacturing, Rapid prototyping and rapid tooling

UNIT-III (08 Lecture)
Grinding & Super finishing:
Grinding: Grinding wheels, abrasive & bonds, cutting action. Grinding wheel specification. Grinding wheel wear - attritions wear, fracture wear. Dressing and Truing. Max chip thickness and Guest criteria. Surface and cylindrical grinding. Centreless grinding.
Super finishing: Honing, lapping and polishing.

UNIT-IV (08 Lecture)
Metal Joining (Welding):
Joining/fastening processes: Physics of welding, brazing and soldering; design considerations in welding, Solid and liquid state joining processes, Survey of welding and allied processes., Gas welding and cutting, process and equipment. Arc welding: Power sources and consumables. TIG & MIG processes and their parameters. Resistance welding - spot, seam projection etc. Other welding processes such as atomic hydrogen, submerged arc, electroslag, friction welding. Weld decay in HAZ.

UNIT-V (08 Lecture)
Unconventional Machining Processes:
Abrasive Jet Machining, Water Jet Machining, Abrasive Water Jet Machining, Ultrasonic Machining, principles and process parameters. Electrical Discharge Machining, principle and processes parameters, MRR, surface finish, tool wear, dielectric, power and control circuits, wire EDM; Electrochemical machining (ECM), Laser Beam Machining (LBM), Plasma Arc Machining (PAM) and Electron Beam Machining.

Books and References:

1. Kalpakjian and Schmid, Manufacturing processes for engineering materials (5th Edition)- Pearson India, 2014.
2. Mikell P. Groover, Fundamentals of Modern Manufacturing: Materials, Processes, and Systems.
3. Manufacturing Technology by P.N. Rao., MCGRAW HILL INDIA.
4. Materials and Manufacturing by Paul Degarmo.
5. Manufacturing Processes by Kaushish, PHI.
6. Principles of Foundry Technology, Jain, MCGRAW HILL INDIA
7. Production Technology by RK Jain.
8. Degarmo, Black &Kohser, Materials and Processes in Manufacturing.

https://www.notebookpresentation.com/